

Comparing Generative Recommendation Systems: Quality, Cost, and Prediction Structure

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Abstract

Web platforms increasingly use generative models to retrieve recommendations from interaction histories, but choosing a system requires understanding how generation strategy, training resources, prediction formulation, and item representation jointly affect its behavior. We organize an empirical comparison of pretrained autoregressive and compact masked-generation recommenders, using existing single-item and two-item experiments across two commerce datasets. The current evidence includes historical performance and resource measurements, prediction-level diagnostics, audited collision-aware evaluation, controlled re-decoding of 18 retained masked-generation checkpoints, and a structural analysis of all 27 identifier maps. Across 18 retained autoregressive configurations, correcting code-level rankings for item ambiguity changes the leading configuration; a verified reassignment also removes catalog collisions while preserving identifier prefixes. With the checkpoints held fixed, confidence-guided reveal order trails a fixed order by 0.95 points of SID Hit@10 at an equal beam of 64 but leads by 0.22 points at 256. The deficit grows with identifier depth, as alternative reveal orders reach the same codes and occupy up to 91% of the final beam. These findings motivate a controlled study with matched compact backbones, common temporal origins, and a fixed three-digit, size-256 representation. We distinguish completed historical and decoding observations from the pending matched comparison and specify focused tests of prediction structure and history use. The study connects recommendation quality and cost to the conditions under which each system is evaluated, without attributing implementation differences solely to the generation paradigm.

CCS Concepts

• Information systems → Recommender systems.

Keywords

generative recommendation, autoregressive models, masked diffusion, multi-item prediction, evaluation, user history

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1 Introduction

Recommendation is a central access mechanism for Web commerce: an interaction history must be converted into a small set of useful catalog items under finite training and serving

budgets. Generative recommenders perform this selection by predicting item representations. Autoregressive models commit to successive tokens, while masked generators iteratively infer missing target components. Their practical comparison depends on more than this distinction: pretraining, model size, supervision, representation, search, and evaluation can all change the apparent result. The Web-related question addressed here is how to evaluate these choices when selecting a recommendation system for catalog discovery. It directly concerns search, recommendation, and personalized access to Web content.

Our central question is: *under which conditions do the tested generative recommendation systems provide useful quality-cost tradeoffs, and what explains their behavior?* Semantic identifiers (SIDs) are one experimental dimension. Varying their construction helps test the stability of a system-level conclusion; it does not define the entire research problem. Prediction structure provides another dimension: recommending the next item, forecasting an ordered pair jointly, rolling a next-item model forward, and predicting each future position independently impose different dependencies and computational costs.

We begin with an existing experimental archive containing pretrained Qwen-based AR and compact DiffGRM-based systems, two commerce categories, and a grid of 27 representations per category. This investment supplies historical results, retained predictions and checkpoints, resource traces, and static catalog analyses. It also exposes important interpretation limits. The models have very different capacities and supervision; historical cohorts and pair metrics are heterogeneous; and full-code collisions can reward identifying an item group rather than the intended item. Consequently, an observed score difference is evidence about the recorded systems and protocols, not by itself a causal effect of autoregression or diffusion.

This working study has four components. First, we consolidate the available performance, resource and prediction diagnostics with their provenance and measurement boundaries. Second, we use collision-corrected evaluation and verified reassignment to examine identity ambiguity as a sensitivity of the comparison. Third, we intervene on decoding alone: retained masked-generation checkpoints are re-decoded under matched search budgets and every fixed reveal order, and each identifier map is analyzed digit by digit against permutation nulls. Fourth, we define a bounded matched-backbone diagnostic, followed by history interventions that can explain a selected behavioral finding. The diagnostic fixes a three-digit, size-256 native representation and reuses the prepared data; wider configurations supply robustness checks rather than a mandatory new factorial sweep.

Evidence status. Historical observations, verified catalog interventions, controlled re-decoding of retained checkpoints, and static identifier-map analyses are reported as completed results. Matched training, independent-position prediction, and neural intervention results remain pending and are marked explicitly in their respective subsections. The present manuscript formalizes the accumulated evidence and the remaining comparisons; it does not claim a universal winning generation paradigm.

2 Related Work

Comparative evaluation of recommendation systems. RecBench compares language-model recommenders with conventional models across item representations and recommendation tasks, jointly examining effectiveness and inference efficiency [7]. The Cold-Starts reproducibility study compares generative systems, including TIGER, OneRec, and DiffGRM, under shared user- and item-cold-start protocols and examines scale, identifiers, and training strategy [15]. GRID provides a modular framework and component ablations for semantic-ID generative recommendation [2]. These studies already establish comparative evaluation, reproducibility, and pipeline decomposition as research directions. Our study compares generative systems in quality, cost, and use of history, with prediction horizon and generation procedure as controlled diagnostics. Conventional recommenders establish whether generation offers a useful tradeoff; comparisons between existing systems and comparisons within a shared compact architecture answer different attribution questions.

Autoregressive and diffusion generation. TIGER generates item identifiers autoregressively from interaction histories [9]; MiniOneRec supplies a framework for pretrained generative recommendation, including supervised fine-tuning and reinforcement learning [6]. DiffGRM combines masked diffusion with parallel semantic encoding, coherent noising, and confidence-guided denoising [8]. MaskGR compares masked SID generation with autoregressive and conventional baselines, including tradeoffs between quality and function evaluations [10]. Its next-two experiment generates pair beams and averages recall across positions. LPDO directly studies ordered future behavior trajectories using a diffusion objective with listwise preference supervision and trajectory-level evaluation [1]. Thus multi-item diffusion recommendation and its efficiency evaluation have direct precedents. We distinguish next-item training, recursive rollout, joint horizon training, and independent future-slot prediction, and assess complete ordered tuples alongside positional scores. Producing two items alone does not establish that a system uses dependencies between them.

Representation and evaluation controls. Semantic IDs are one representation choice within this comparison. Wang et al. [12] connect SID neighborhoods with item selection and study how early decoding decisions restrict later candidates. EPIC conditions denoising on explicit posterior evidence over compatible catalog items, including analyses of history and

partial-code ambiguity [11]. These works motivate controlling the surviving catalog candidates when interpreting generation behavior. Zhang et al. [14] introduce Collision-Corrected Evaluation (CCE) and Zero-Collision Reassignment (ZCR). We adopt both as established methods: CCE corrects next-item credit under ambiguous identifiers, whereas ZCR changes the assignment and consequently requires retraining for a recommendation comparison. Neither method is a contribution of this study. Representation sensitivity and collision correction support the system comparison; static codebook statistics alone cannot establish a recommendation advantage or a learned mechanism.

Explaining system differences. Sparse-autoencoder studies interpret and control sequential-recommender features [4] and diffusion-language-model activations [13]. Comparisons of autoregressive models with their masked-diffusion adaptations examine computational changes through circuit analysis and ablations [5]. The Confidence Shortcut further studies interactions between masking coverage and confidence-based decoding on controlled reasoning tasks [3]. These findings motivate tests rather than establish mechanisms in recommendation. Our supporting interpretability arm separates catalog structure, information recoverable from activations, and effects of history or internal interventions. In particular, history sensitivity is measured while controlling visible target information and compatible candidates. Such evidence can help explain an observed system difference; attributing it to the generation paradigm also requires controls for architecture, pretraining, supervision, and search.

3 Preliminaries

3.1 Recommendation Tasks and Prediction Protocols

Let $\mathcal{I}$ denote the catalog, $H_{u,t} = (i_{u,0}, \dots, i_{u,t-1})$ the chronological history available at origin t , and $Y_{u,t} = (i_{u,t}, \dots, i_{u,t+h-1})$ the ordered future target. We study $h = 1$ and $h = 2$. Forecasting ordered future events differs from selecting items for presentation as a recommendation slate: exchanging future positions changes the target.

Next-only trains and evaluates a single future item. *Rollout* applies a next-item model repeatedly, appending each predicted item to the history before predicting the following item. *Joint* prediction trains on the full horizon and permits dependence between future positions; it can use autoregressive or masked generation. An *independent-slot* control predicts each future position from the same history without access to other future predictions. At the item-distribution level its factorization is $P_{\text{ind}}(i_1, i_2 \mid H) = P_1(i_1 \mid H)P_2(i_2 \mid H)$. The slots may share parameters; independence concerns their conditioning. The main evaluations supply no observed future item; oracle conditioning is a separately labeled diagnostic. Conventional catalog scorers share the corresponding targets and ranking metrics.

3.2 Identifiers and Generative Scores

A semantic-ID map $s : \mathcal{I} \rightarrow \{0, \dots, B-1\}^D$ represents an item by D position-specific codewords, as in generative retrieval [9]. Concatenating h codes yields target tokens $y = (y_1, \dots, y_{hD})$. The inverse group $G(c) = \{i \in \mathcal{I} : s(i) = c\}$ may contain multiple items. A correct complete SID then identifies a group, not necessarily the target item. The fraction of items in non-singleton groups differs from duplicate excess $1 - |s(\mathcal{I})|/|\mathcal{I}|$.

An autoregressive model minimizes negative log likelihood under

$$p_\theta(y | H) = \prod_{j=1}^{hD} p_\theta(y_j | H, y_{<j}). \quad (1)$$

A masked model predicts from H and a partially revealed target $\tilde{y}$. In the current DiffGRM-derived implementation, n indexes supervised views and m_{nj} indicates a masked position:

$$\mathcal{L}_{\text{mask}} = \frac{\sum_{n,j} m_{nj} \text{CE}_\epsilon(q_\phi(\cdot | H_n, \tilde{y}_n)_j, y_{nj})}{\sum_{n,j} m_{nj}}, \quad (2)$$

with smoothing $\epsilon = 0.1$ and nonempty masks. Each masked position contributes equally. Iterative reveals change the conditioning information; parallel outputs alone do not demonstrate learned dependence. Search budgets and catalog constraints also affect returned rankings. Our current diffusion decoder uses a heuristic path score, not a normalized joint probability; AR beam scores likewise do not establish catalog-normalized item probabilities.

3.3 Reveal Orders and Beam Search

A masked decoder completes the D positions of one target SID in D steps. A *fixed* policy reveals positions in a declared permutation π of $\{1, \dots, D\}$. A *confidence* policy lets every beam candidate reveal any still-masked position, ranking the expansions by accumulated log-probability of the revealed codes. With beam cap B , both policies keep the B highest-scoring partial candidates after each step, discard complete SIDs absent from the catalog, and deduplicate complete SIDs by maximum score. Under the confidence policy, different reveal orders can reach the same partial assignment, and the beam keeps each such path. An equal cap B therefore implies neither an equal number of distinct candidates nor equal decoder work.

3.4 Ranking and Collision Correction

For exact target rank r , $\text{Hit}@K = \mathbf{1}[r \leq K]$ and $\text{NDCG}@K = \mathbf{1}[r \leq K] / \log_2(r+1)$, both zero when absent. For $h=2$, the ranked object is a complete ordered tuple. Positional hits are separate diagnostics and need not imply a tuple hit.

We adopt Zhang et al.'s next-item CCE [14]. For distinct ranked valid codes $(c_1, \dots, c_R)$, let $g_r = |G(c_r)|$ and $a_r =$

$\sum_{j < r} g_j$. If the target belongs to $G(c_r)$, its expected item-level credit under uniform ordering within that group is

$$\text{ItemHit}@K = \frac{\min\{g_r, \max(0, K - a_r)\}}{g_r},$$

$$\text{ItemNDCG}@K = \frac{1}{g_r} \sum_{\ell=1}^{g_r} \frac{\mathbf{1}[a_r + \ell \leq K]}{\log_2(a_r + \ell + 1)}. \quad (3)$$

Here K counts expanded item positions. Uniform credit is an evaluation convention for unresolved identity, not learned discrimination. Singleton groups recover standard metrics. We do not extend this formula to pairs: ordered-SID-tuple scores and collision strata do not identify colliding item pairs.

3.5 Representation and Explanatory Controls

ZCR [14] preserves each code's first $D-1$ digits and reassigns the final digit to make items distinct. With fixed centroids, it first minimizes changed assignments, then quantization cost. Prefix groups must contain at most B items. Unlike CCE rescoring, ZCR changes histories and targets, requiring retraining for quality comparisons.

For visible positions V , compatible items are

$$A(V, s_V(i)) = \{i' \in \mathcal{I} : s_d(i') = s_d(i) \ \forall d \in V\}. \quad (4)$$

Matching visible positions and values controls this support when probing history dependence; matching cardinality alone does not. Catalog ambiguity, decodable activation information, and behavioral effects of interventions constitute distinct evidence.

4 Comparative Study Design

Our central question is how generative recommendation systems differ in prediction quality, computational cost, and use of interaction history. Semantic-ID construction is a controlled experimental factor. We distinguish two levels of comparison: the practical behavior of the retained pretrained AR and compact masked systems, and prospective experiments that align their backbones and training information. The latter experiments are planned; historical measurements do not establish their outcomes.

Questions and scope. We ask whether system comparisons persist under common event cohorts and item-aware evaluation; whether predicting two future interactions jointly offers an advantage over independent prediction or sequential roll-out; and which history information explains the observed behavior. Quality and cost are evaluated together. A difference between the existing Qwen and DiffGRM implementations cannot by itself be attributed to autoregression or masked prediction, because their scale, initialization, objectives, and input representations differ. Likewise, prediction of two observed future events does not establish planning ability or the utility of a displayed slate.

A bounded matched pilot. The primary prospective anchor is the native RQ-VAE 3×256 map on Industrial, followed by confirmation on Office. We hold this map fixed across two

compact generators sharing history encoding, dimensions, and training information. Matching modules will start from identical parameter tensors for each seed. Four initial trainings cross generator type (causal AR or masked prediction) with target objective (next-item only or joint next-two). The next-item models also supply a rollout baseline: generate the first item, append the prediction to history, and generate the second. This adds an inference condition without another training run. All conditions rank complete ordered target tuples with explicitly recorded scoring rules and search budgets.

The pilot uses the same horizon-two origins and boundaries for every condition. A next-item-only model receives the first label at each of those origins; it does not revert to the usual horizon-one split. Otherwise, the standard next-item validation event would coincide with the first horizon-two test target. Checkpoint selection and tuning use only the reserved validation block, with the same declared pair-ranking endpoint. An oracle first item may be supplied in a separate diagnostic, but cannot replace the generated first item in the rollout result.

Two subsequent trainings, one per generator, will test independent future prediction. A horizon-conditioned one-item decoder predicts each future position from the same original history, without access to the other target. Its two rankings are combined using a declared tuple-scoring rule. This distinguishes joint target dependence from merely supervising a second future event. The four-run pilot and six-run extension each describe one seed; replication multiplies those counts.

Representation and explanatory controls. The primary native map can retain collisions, so we report SID-tuple accuracy and singleton-target strata without claiming item identity for ambiguous codes. A separate RQ-KMeans 3×128 native/ZCR stress test can add four trainings: two generators under two assignments, holding the next-item objective and raw cohort fixed so that CCE remains applicable. It measures assignment sensitivity without substituting a different map into a trained model. A next-two stress extension is optional and separate. Wider SID sweeps remain supporting evidence rather than the organizing contribution.

History interventions will begin with fixed candidate pairs and matched visible target states. Relevant and irrelevant history replacements, fixed-candidate scoring, and free decoding separate changes in model response from search pruning. Selective activation restoration or disruption will follow only a reproducible behavioral effect. Reduced history use after the catalog fixes a target is not, by itself, a failure. Confirmation will report training-seed variation and paired user uncertainty; a static semantic map or a decoded feature alone does not establish causal use.

5 Experimental Setup

Data and provenance. The retained inventory contains the Industrial and Scientific and Office Products Amazon categories. The audited Industrial cache contains 6,297 users, 3,105 items, and 43,102 interaction events. Dataset labels

in the loader and directory names do not establish the upstream release or filtering procedure. Office contains 61,564 users, 17,696 items, and 472,091 events; its cache matches the manifest-verified interaction and identity files. Transferred interaction, metadata, identity-map, and SID files are tracked by content hashes; final reporting still requires verification of the original release and embedding construction. Original fitted quantizer states are unavailable for several supplied maps.

The maintained v3 protocol reserves zero-based positions $[L - h, L)$ for test and $[L - 2h, L - h)$ for validation. Training target blocks end by $L - 2h$. Every origin requires at least two observed events, and histories retain the most recent 50. Sliding training windows may overlap within training; target events cannot cross split boundaries. This is a within-user chronological holdout, not a global calendar-time split. Repeated products remain valid interactions. Industrial has 17,914 training examples and 6,297 examples in each held-out split for next-item prediction. Its next-two split has 5,758 training, 3,624 validation, and 6,297 test examples. Office has 225,835 next-item training examples and 61,564 in each held-out split; next-two has 98,261 training, 39,333 validation, and 61,564 test examples. These are replayed cohort counts, not completed performance comparisons. The planned matched pilot uses horizon-two origins for all objectives (Section 4).

System differences. Qwen undergoes full fine-tuning on concatenated SID recommendation, item-title/SID mapping, fused-history recommendation, title recommendation, and title-history-to-SID datasets. Their labeled-token exposure is unequal. Training retains the final 512 tokens of prompt plus response, so shared raw histories do not guarantee shared effective histories. Diff compresses each history item’s code embeddings into one vector and uses guided masked cross-entropy with label smoothing 0.1. With $T = hD$ target positions, its current automatic schedule constructs T supervised views per example, covering one through T masks, plus a confidence-scoring pass. Raw batch size therefore does not equal decoder workload. The historical Qwen evaluator also words the SID history differently from the SID-to-SID training data (“...given the following chronological interaction history: ...” versus “The user has interacted with items ... in chronological order.”); every archived AR prediction uses the evaluation wording. Reconstructed evaluator inputs match the upstream token sequences for every test row of the three retained AR checkpoints. Test-file row order matches all 28 archived AR prediction files, which carry no user identifiers and can be joined to histories only by that order.

The current implementations also require alignment before the matched pilot: proportional warmup, consistent projected cross-attention, explicit history-padding masks, and target-slot information for revealed codes. In particular, the inherited Diff warmup is 10,000 optimizer updates; Industrial next-two at batch 256 provides only 4,600 updates even over 200 epochs. The proposed control will specify and verify these settings before training, retain full-mask supervision, and use

Table 1: Recorded maintained systems and the planned matched control. Parameters exclude upstream item encoders and quantizers. Current defaults are not claims about every historical run. The compact control has not been trained; its final count follows the shared slot-encoding changes.

System	Architecture	Parameters	Full-run defaults	Evidence status
Qwen AR	28 decoder layers; width 1,536; pretrained Qwen2.5-1.5B-Instruct	About 1.544B at the primary anchor	Batch 1,024; LR 3×10^{-4} ; at most 10 epochs; validation loss	Historical full runs; bounded revised-protocol checks
DiffGRM	1 encoder/4 decoder layers; width 256; trained from scratch	5.37–6.06M across the retained grid	Batch 256; configured LR 0.003; at most 200 epochs; tuple NDCG	Historical full runs; bounded revised-protocol checks
Matched AR/masked	Shared 1 encoder/4 decoder layers; width 256; MLP 1,024; 4 heads	Approximately 5.47M each at 3×256	Shared cohort, tuning allowance, validation endpoint, and recorded compute budget	Planned; no comparative results yet

identical shared parameter initialization and input information. These are comparability requirements, not completed experimental interventions.

Controlled decoding protocol. We re-decode the 18 retained next-item Diff checkpoints (RQ-VAE, RQ-KMeans and MQ at depths three to five and widths 128 and 512) on the archived last-event cohort of 6,297 Industrial users. The decoder applies each checkpoint’s projected cross-attention cache. Before any new condition was scored, it reproduced the archived Hit@10 and NDCG@10 of the fixed-order ($B = 64$) and confidence ($B = 256$) decoders for all 18 checkpoints, with a maximum absolute difference of 8.5×10^{-10} . Exact reproduction required FP32 arithmetic without TF32, batch size 32, and decoder chunks of 1,024. In the shared search, both policies use the same beam cap, full expansion of every digit, score accumulation, deterministic tie handling, final catalog filtering, and maximum-score SID deduplication; only the position a candidate may reveal next differs. The archived confidence decoder, which fills its last digit greedily, is reported separately. The all-order experiment was specified before inference: six depth-three checkpoints, all six permutations, $B = 64$, and confidence minus the mean fixed-order SID Hit@10 as the primary estimand. Intervals use 2,000 paired bootstrap draws of users that keep every checkpoint and condition of a user together. They exclude training-seed variation, and the checkpoints were originally selected with confidence-guided validation. Item-uniform Hit@10 applies Eq. (3) to the diffusion rankings.

Identifier-map analyses. Structural analyses use the 27 frozen Industrial maps, the 2,560-dimensional item embeddings supplied with them, and level-1 categories of the Amazon-2018 category hierarchy joined by item identifier. These labels were acquired after training and serve interpretation only. Standalone variance is the between-group share of embedding variance for the partition induced by one digit. Category adjusted mutual information (AMI) is computed for a digit alone and within its parent prefix. Conditional estimates are shown only when at least eight labeled items

per parent, 100 items, and five parents contribute. Purity is the majority-category share among items in multi-item prefix groups, compared with 200 label permutations within the same groups. The refinement index is $1 - R_w^2/\mathbb{E}[R_w^2]$, where R_w^2 is the within-group share of embedding variance and the expectation $(n - k)/(n - 1)$ is taken over random partitions of the n items into the same number k of groups. These quantities describe the maps, each from one fit, not trained models.

Evaluation and uncertainty. The maintained next-item evaluation reports exact-SID and CCE item metrics. Revised next-two results use exact ordered-SID-tuple Hit/NDCG, separately named slot diagnostics, and collision strata; CCE is not extrapolated to pairs. Both decoders use catalog constraints and approximate search. AR scores include boundary tokens; Diff uses a fixed-order masked reveal-path score, which is not a normalized joint likelihood. Historical Diff next-two checkpoints were selected using overlapping validation/test target events, and require fresh training for the revised holdout. Historical AR prediction cohorts are likewise kept separate from new shared-cohort comparisons.

We will record wall time, GPU count and type, precision, peak memory, optimizer updates, supervised target exposure, and decoder calls. Inference timing must include catalog operations and rescoring, with batch size and returned ranking length controlled. Equal beam width does not imply equal cost. Three matched seeds are the intended confirmatory starting point, subject to pilot variability and available compute. Within fixed saved predictions, paired bootstrap intervals resample users across conditions; they do not replace training-seed replication. Bounded execution checks remain excluded from recommendation-quality tables.

6 Experiments and Analysis

6.1 Historical Evidence and Its Scope

The existing experiments provide a broad descriptive record, while the controlled comparison remains incomplete. The reconciled workbook contains 221 observations, including

Table 2: Workbook snapshot coverage: completed SID settings out of 27 per cell. Industrial AR next-item includes one ambiguous setting; later AR pair recovery is reported separately below.

Dataset	Task	AR	Diff family
Industrial	Next item	27	27
	Next two	10	22
Office	Next item	9	27
	Next two	0	0

planned rows, of which 127 have all six reported ranking metrics. Its core grid has 121 complete, unambiguous observations; four additional model variants and two conflicting observations of one nominal baseline are retained separately. Figure snapshots, current run records, and collision-corrected evaluations are distinct evidence sources. The following analyses and the accompanying result inventory account for every available family without treating missing cells as zeros.

The accompanying result inventory and browsable supporting archive retain all 44 distinct existing result figures, their numerical tables, the 27 current completed AR pair records, and source hashes. These are dated snapshots, not a pooled leaderboard. Repeated shareable figures and the combined figure PDF are duplicate presentations; missing metrics remain missing. No raw user histories or prediction dumps are included in this compact archive.

All historical comparisons below concern the recorded systems. They do not isolate autoregression from diffusion: model size, pretraining, supervision, cohorts, selection, and decoding differ. In particular, historical Diff next-two validation and test targets overlap, and selected checkpoints inherit that problem. Their aggregate scores and decoder ablations remain visible as archival diagnostics, not independent held-out evidence. No training-seed uncertainty is available for these comparisons.

6.2 Dataset Dependence of the Recorded Comparison

Across 26 unambiguous Industrial next-item SID settings, the workbook’s mean diffusion-minus-AR gap is -1.784 percentage points in reported HR@10 and -1.272 in NDCG@10. Across nine Office settings, the respective gaps are $+6.650$ and $+5.028$. Restricting both datasets to the same eight width-256 settings preserves the difference in direction (Figure 1). This motivates dataset replication; it does not establish a universal winner or eliminate the training and cohort differences between the archived systems.

SID depth is another associated factor. Industrial mean HR@10 gaps are -0.37 , -1.07 , and -3.76 points at depths three, four, and five; Office remains positive at each depth. An inspected historical depth-five Diff training path lacks a supervised fully masked view, so the pattern cannot be attributed to sequence length alone. Search adds a further confound. For the 18 retained width-128/512 Diff checkpoints,

whose archived scores our decoder reproduces exactly, the archived confidence decoder at its beam of 256 returned ten distinct legal SIDs to 99.96%, 98.6%, and only 77.1% of users at depths three, four, and five (Section 6.6). Figures 3 and 2 retain the depth analysis and model variants, including the two conflicting RQ-VAE 3×256 AR scores, 11.78 and 12.85, without selecting one.

Model and SID variants. The conflicting Qwen2.5-1.5B rows lack distinguishing training labels, so neither anchors a core comparison. The workbook groups “Mask Diffusion” and “DiffGRM” under one family; this does not establish identical architecture or execution. All reported cutoffs, collision associations, and coverage grids remain in the archive. The earlier collision gallery also preserves within-tokenizer associations, factorial marginal means, metric agreement, and artifact-sensitivity analyses. These are observational configuration summaries, not causal effects or adjusted leaderboards. Workbook collision percentages can denote duplicate excess and include stale width-256 values; the audited CCE tables use the distinct item-in-collision-group definition.

6.3 Training and Evaluation Cost

The efficiency archive contains 62 completed-run records: 28 AR and 34 DiffGRM. For the 28 AR records, evaluation accounts for 3.9–5.6% of reported training-plus-evaluation GPU-hours for next-item prediction and 39.2–49.7% for next-two prediction (Figure 4). Reported evaluation time is 45.1–58.1 ms/example and 646.6–965.8 ms/example, respectively. These are job-level amortized times, not batch-one latency; tasks also have different SID coverage and cohorts.

DiffGRM’s records instead measure whole Slurm jobs on one GPU whose model is unspecified. Their timing cannot be normalized against the AR records to claim a paradigm speedup. The archive also retains utilization, memory, power, sampled energy, and estimated training compute for AR; comparable Diff fields are absent. Failed and interrupted attempts are excluded from these completed-run totals. The historical accuracy–cost plots join nominal configurations without a shared run identifier and are therefore descriptive, not verified execution-level Pareto frontiers.

6.4 What the Historical Pair Predictions Recover

The legacy pair metric assigns relevance 1 when both SID slots match, 0.5 when exactly one matches, and 0 otherwise. Its reported HR takes the maximum relevance among the first K pairs; its NDCG discounts the first occurrence of that maximum. These quantities differ from exact ordered-pair Hit/NDCG. In the workbook’s ten matched pair settings, mean diffusion-minus-AR HR gaps change from $+0.57$ at $K = 3$ to -1.76 at $K = 10$, while the NDCG@10 gap is $+0.025$. Overlapping Diff holdouts and historical decoding prevent a clean performance conclusion.

The saved AR beams support a more specific observation (Figure 5). Across nine width-256 configurations, first-slot SID recovery at ten ranges from 12.98% to 22.60%, versus 5.74%

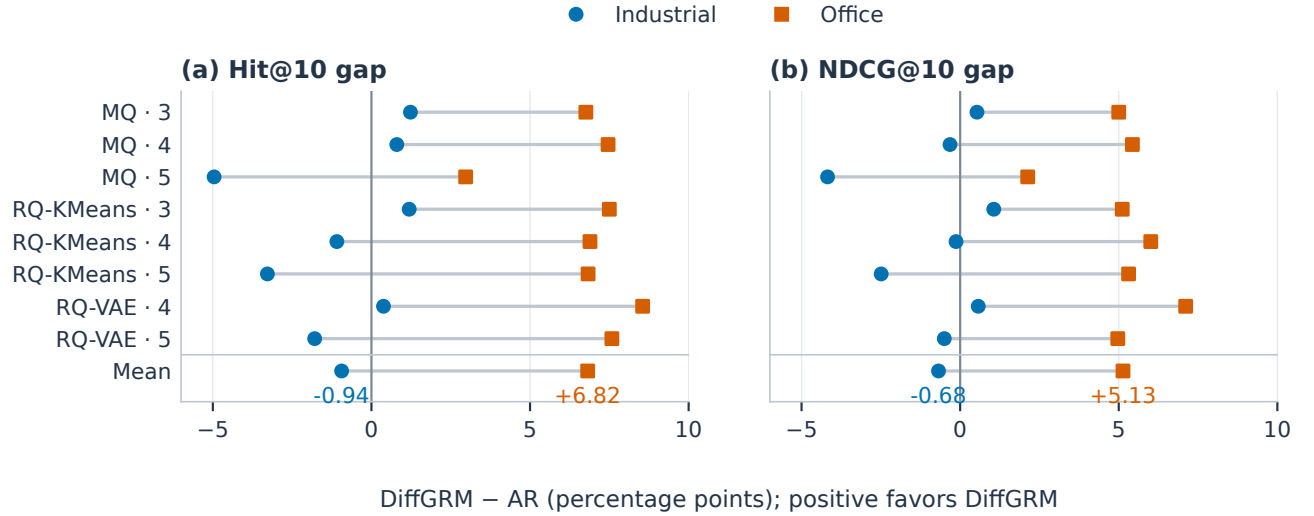


Figure 1: Historical next-item comparison at eight identical SID settings in Industrial and Office. Positive gaps favor the workbook’s diffusion family. Nominal SID matching does not match training or evaluation protocols; reported metrics are not collision-corrected. The ambiguous Industrial RQ-VAE 3×256 AR baseline is excluded from both sides.

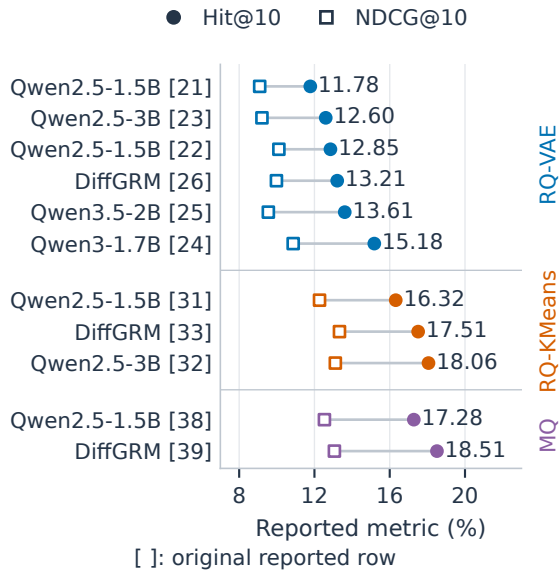


Figure 2: Recorded Industrial width-256, depth-three model variants and both ambiguous baseline rows. These are not a controlled scaling curve.

to 9.99% for the second slot. Exact ordered SID-pair recovery is only 1.80–2.93%. Recovering both target SIDs somewhere in a list need not recover their ordered pair. These differences motivate separate slot, tuple, and rollout diagnostics rather than a single graded score.

Second-slot recovery is associated with first-slot recovery, but the archive cannot show that it depends on it. For the

top-ranked pair, the equal-configuration probability that the second SID is correct is 4.03% when the first is correct and 0.34% when it is wrong (difference +3.69 points; user-cluster 95% interval [+1.14, +6.23]; ten configurations, 34,520 rows). The association is heterogeneous: five of ten configuration estimates are positive, ranging from -0.55 to $+18.48$ points, and conditioning on first-slot correctness selects easier examples. Ordered target pairs are recovered only marginally more often than their reversals (Hit@20 difference +0.23 points with equal configuration weights). The saved AR pairs have no scores and no controlled first-item interventions, so they cannot establish that the second prediction uses the first.

The historical decoder returns 17.7–19.6 of 20 requested pairs on average, with distinct first SIDs across its candidates. This constrains exploration of alternative continuations for the same first prediction. Seventeen additional AR runs completed during recovery; together with ten earlier runs they cover all 27 Industrial SID settings. Their scores are listed in Table 3. They retain historical cohorts and graded metrics; completion does not convert them into revised-protocol results or update the older figures retroactively.

Decoder and training diagnostics. The AR beam archive records first/second-slot recovery, exact and reversed SID pairs, prefix agreement, and returned-list yield. At rank one, 75–92% of predictions share no leading SID digit with the first target. Single-item AR beams show the same front-loading. Across the 18 audited next-item configurations (66,258 example–configuration rows; 1,606 users), the first wrong digit of the rank-one SID is the first digit in 82.16% of rows (user-cluster interval [80.84, 83.48]), and exact rank-one SID accuracy is 8.75%. Once a prefix is correct, the next digit is wrong in 35.89% of surviving rows at digit two and

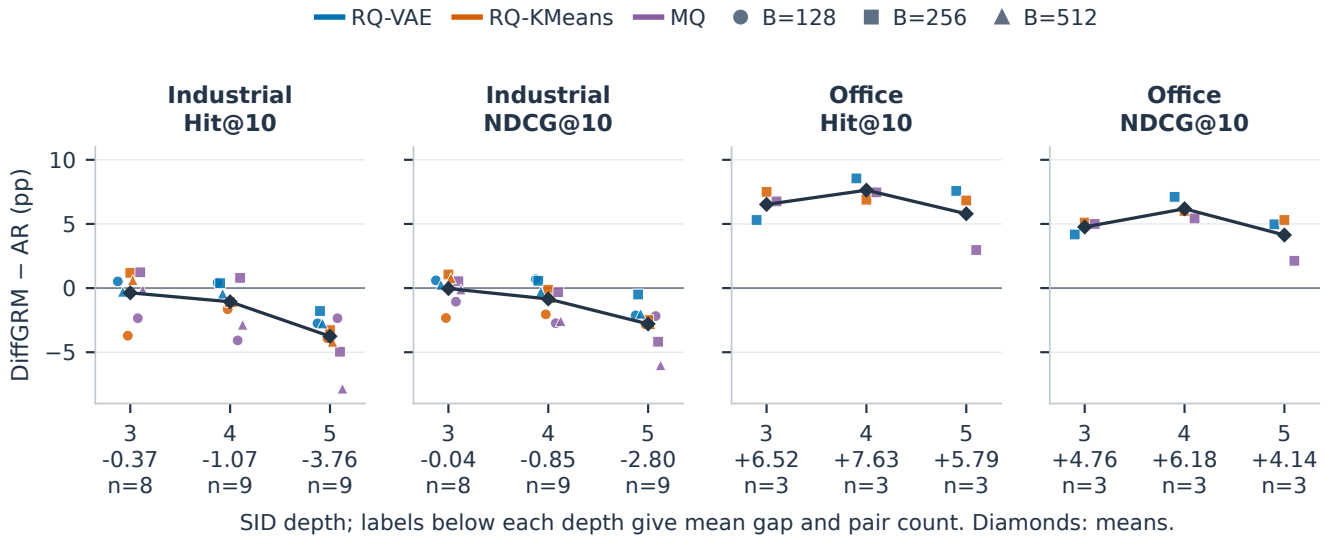


Figure 3: Historical gaps by depth. Points are configurations, not seeds; coverage differs between datasets. The depth-five Diff masking-coverage confound prevents a sequence-length attribution.

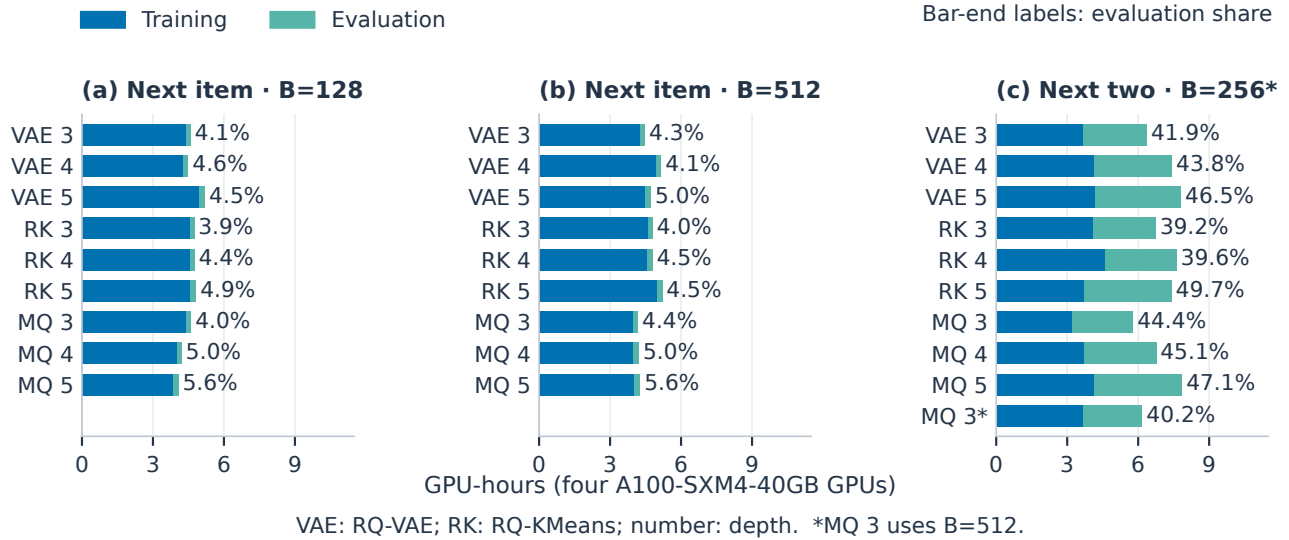


Figure 4: Historical Qwen2.5-1.5B training and evaluation allocation on four A100-40GB GPUs. Each bar is a completed configuration. The larger evaluation share for the recorded pair runs motivates measuring decoding cost explicitly, but the unmatched task settings do not identify a task-only slowdown or an AR-versus-Diff speedup.

17.98% at digit three. This describes survivorship in an autoregressive error process, not independent digit difficulty. The historical title-compatible Hit@10 (16.72%) slightly exceeds exact-SID Hit@10 (16.59%) because of its legacy title match. Comparing first-slot and standalone next-item scores uses different cohorts (3,452 versus 3,681 examples), so it cannot establish that pair training preserves next-item quality. Among seven completed historical Diff decoder-order comparisons, six NDCG@10 changes are at most 0.17 points

in magnitude; RQ-VAE depth four changes by 1.44 points. These inherit the old split/decoder limitations. AR validation cross-entropy, Diff masked training loss, and leaked Diff validation relevance curves are preserved as separate diagnostics, never placed on a common loss scale.

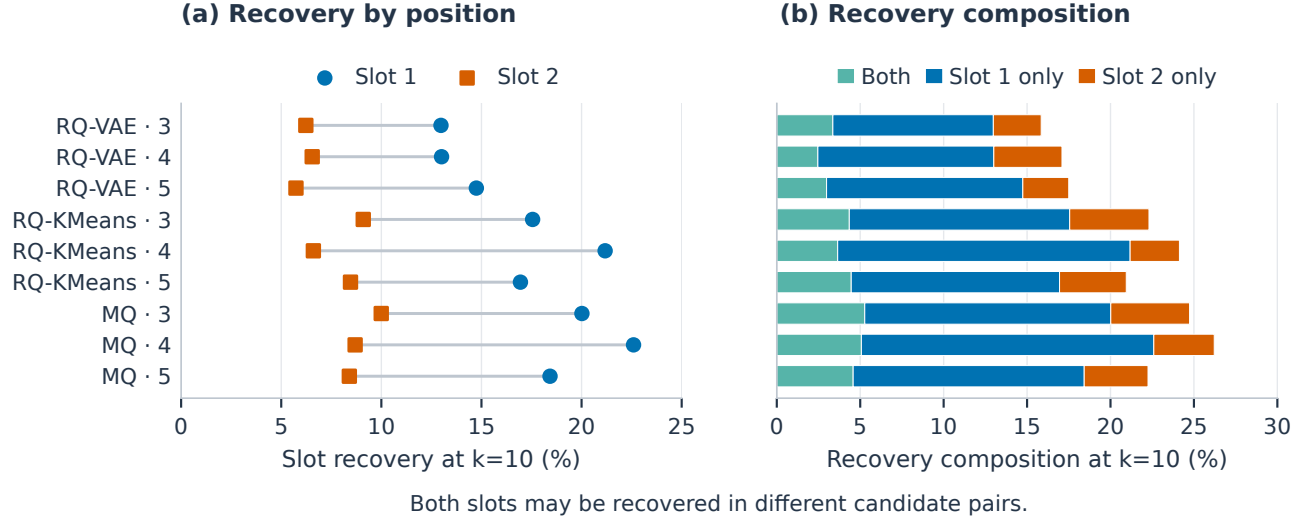


Figure 5: Historical AR-only pair diagnostics on 3,452 examples per configuration. Slot recovery denotes SID matches; collisions can leave item identity unresolved. In panel (b), “both somewhere” may use different candidate pairs. These are not controlled next-item-versus-pair comparisons or measurements of Diff behavior.

Table 3: All 27 completed historical Industrial AR pair cells as of 21 September. Entries are reported graded HR@10/NDCG@10 (%). Superscript r marks the 17 recovery runs; unmarked cells predate recovery. Neither metric is exact-pair accuracy.

Family	D	$B = 128$	$B = 256$	$B = 512$
RQ-VAE	3	9.28/6.39 ^r	9.02/5.93	9.78/6.50 ^r
	4	8.15/5.78 ^r	9.44/5.73	6.91/3.75 ^r
	5	8.73/5.31 ^r	9.89/6.41	11.40/7.77 ^r
RQ-KMeans	3	16.18/11.73 ^r	12.60/7.99	13.33/8.66 ^r
	4	14.31/9.90 ^r	13.09/8.80	12.30/8.75 ^r
	5	14.18/9.83 ^r	11.62/7.69	13.85/10.65 ^r
MQ	3	15.82/10.72 ^r	13.83/10.21	13.72/8.80
	4	12.78/8.38 ^r	14.41/10.89	17.06/12.78 ^r
	5	12.20/8.60 ^r	12.33/8.44	12.57/7.74 ^r

6.5 Collision Correction and Reassignment

The strongest independently recomputed evaluation result is the historical next-item CCE audit: 18 AR configurations, each with 3,681 examples from 1,606 users. Applying the existing CCE method [14] changes RQ-KMeans 3×128 from 24.0152% SID Hit@10 to 12.6377% corrected item Hit@10. RQ-VAE 5×512 has the highest observed corrected item scores: 14.1538% Hit@10 and 9.7369% NDCG@10 (Figure 6; Table 4). This is a measurement-dependent ordering, not a new correction method or a multi-seed tokenizer ranking.

The median corrections are 4.127 and 3.537 percentage points for Hit@10 and NDCG@10. After documented trailing-padding removal, 3.74% of rankings contain fewer than ten SIDs; all remain in the denominator. The CCE audit contains no qualifying width-256, Office, Diff, or pair result.

An independent reimplement over the frozen predictions, expanding each ranked SID into its item group with uniform credit, reproduces all 18 corrected Hit@10 values

exactly. The mean correction is 4.543 points (user-cluster interval [4.1, 5.0]). The correction is 14.1 points for targets in collision groups and 1.2 points for singleton targets; the latter arises because earlier predicted groups consume expanded positions. Within configurations, targets in collision groups have 42.5 points higher SID Hit@10 than singleton targets (interval [40.0, 44.9]; positive in 17 of 18 configurations). This is consistent with training labels pooling onto shared codes, not with item identification.

Complete collision-corrected table. Table 4 retains every eligible historical CCE configuration. Correlations and rankings across this fixed grid are descriptive; user-level uncertainty would not substitute for training-seed replication. Native/ZCR smokes, checkpoint reload checks, and source/hash audits remain documented separately from recommendation results.

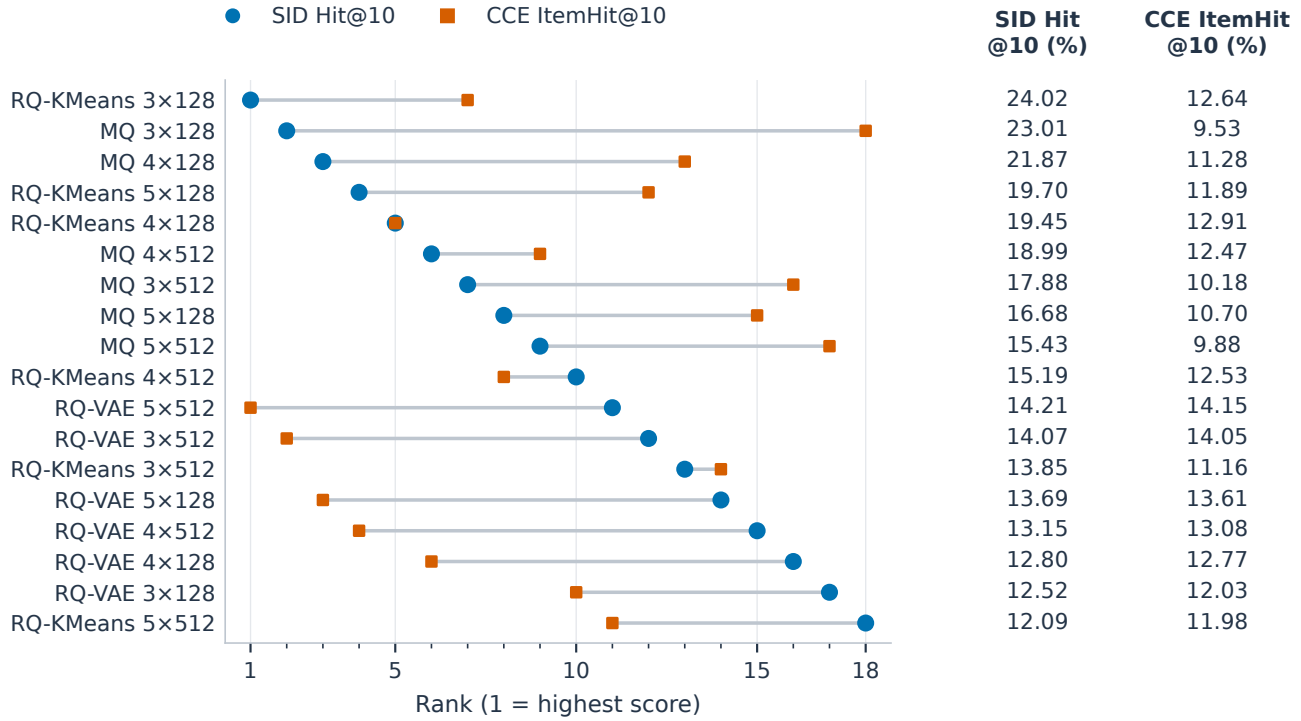


Figure 6: Audited historical AR ranking before and after single-target CCE. One retained run per configuration; widths 128 and 512 only. SID ranks and expanded item ranks have different meanings.

Verified ZCR preserves every prefix and the 384 SID tokens while changing 29.31% of item tails; distortion increases 4.598% (Table 5). Reconstructed geometry reproduces every native assignment, without establishing equality to unsaved original centroids. No full downstream native/ZCR comparison is available.

The planned native RQ-VAE 3×256 anchor has 3,072 distinct SIDs for 3,105 items, with 58 items in collision groups. Under the shared next-two holdout, 580 of 6,297 test pairs touch a colliding target (9.21%). Thus, width 256 provides a practical fixed configuration but does not remove the need for identity-aware reporting.

6.6 Controlled Re-decoding of Retained Diff Checkpoints

The archived comparison of Diff reveal policies paired confidence decoding at $B = 256$ with the seed-42 fixed order at $B = 64$: +0.34 points of SID Hit@10 on the 18 retained width-128/512 next-item checkpoints. We re-decoded the same checkpoints for the same 6,297 users with matched beam caps and a shared search (Section 5); only the position a candidate may reveal next differs between conditions. These are interventions on decoding alone, conditional on checkpoints selected by confidence-guided validation.

The confidence advantage depends on the search budget. At $B = 64$, confidence decoding trails the fixed order by 0.95

points of SID Hit@10; at $B = 256$ it leads by 0.22 (Table 6). Raising the cap from 64 to 256 improves confidence decoding by 1.26 points and the fixed order by only 0.08. Item-uniform Hit@10 agrees in direction: $-0.24 [-0.34, -0.15]$ at $B = 64$ and $+0.34 [+0.24, +0.43]$ at $B = 256$. The $B = 64$ deficit concentrates at larger depths. Mean differences are +0.38, -0.68 , and -2.55 points at depths three, four, and five, against +0.39, +0.46, and -0.17 at $B = 256$ (Figure 7a).

Duplicate paths consume the beam. Several reveal orders can reach one partial assignment, and the beam keeps every path (Section 3.3). At $B = 64$, duplicate paths make up 57%, 79%, and 91% of the confidence policy’s final beam at depths three, four, and five, and only 4.9% of users receive ten distinct legal SIDs at depth five (Figure 7b,c). The fixed order has no duplicate paths but loses 30%, 47%, and 50% of its final paths to SIDs absent from the catalog. Even at $B = 256$, the archived decoder returns ten distinct legal SIDs to only 77.1% of users at depth five. Candidate diversity is therefore a plausible explanation for the budget dependence, not a demonstrated mechanism. The direct test, deduplicating partial states before pruning, remains pending.

All fixed orders at depth three. One seed-42 permutation per depth may be unrepresentative. On six depth-three checkpoints (three quantizers, widths 128 and 512), confidence decoding at $B = 64$ exceeds the mean over all six fixed orders by 0.40 points of SID Hit@10 and 0.29 of NDCG@10

Table 4: All audited historical Industrial next-item AR configurations. Scores and collision rates are percentages. Each row uses 3,681 examples from 1,606 users and one retained training run. These are historical cohort results, not the revised v3 comparison. Collision rate counts catalog items in non-singleton complete-SID groups.

SID family	$D \times B$	SID Hit@10	CCE ItemHit@10	SID NDCG@10	CCE ItemNDCG@10	Collisions (%)
Parallel MQ	3×128	23.0101	9.5287	15.6182	5.9951	54.52
Parallel MQ	3×512	17.8756	10.1755	13.2147	6.7501	30.72
Parallel MQ	4×128	21.8691	11.2760	15.7651	7.0983	37.00
Parallel MQ	4×512	18.9894	12.4678	14.7332	7.9980	22.67
Parallel MQ	5×128	16.6802	10.6992	12.8653	7.1410	25.89
Parallel MQ	5×512	15.4306	9.8755	11.0820	6.7455	18.45
RQ-KMeans	3×128	24.0152	12.6377	17.0918	7.8677	44.80
RQ-KMeans	3×512	13.8549	11.1555	10.1675	7.4302	16.04
RQ-KMeans	4×128	19.4512	12.9069	15.3237	7.9801	29.31
RQ-KMeans	4×512	15.1861	12.5302	11.1184	8.6024	9.66
RQ-KMeans	5×128	19.6957	11.8903	14.5204	7.8097	21.77
RQ-KMeans	5×512	12.0891	11.9804	9.3737	9.3265	0.64
RQ-VAE	3×128	12.5238	12.0295	9.2964	8.3490	1.55
RQ-VAE	3×512	14.0723	14.0451	10.1643	9.3891	0.77
RQ-VAE	4×128	12.7954	12.7683	9.4840	8.7204	0.77
RQ-VAE	4×512	13.1486	13.0807	9.9475	9.0635	0.71
RQ-VAE	5×128	13.6919	13.6104	10.2760	9.4409	0.71
RQ-VAE	5×512	14.2081	14.1538	10.6195	9.7369	0.71

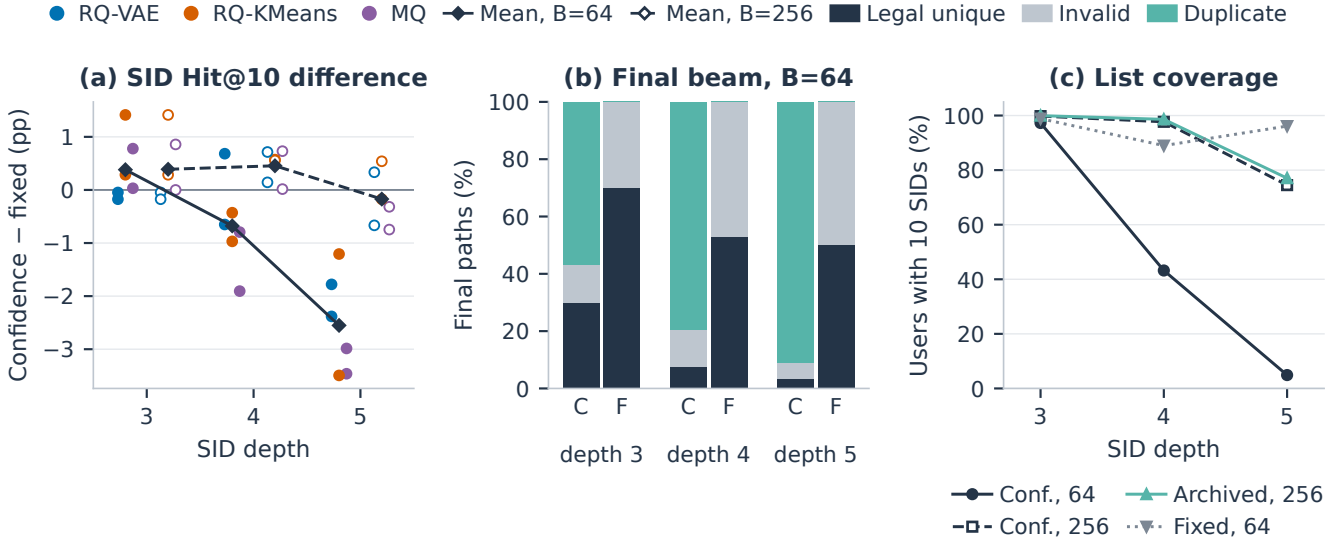


Figure 7: Controlled re-decoding of 18 retained Industrial next-item Diff checkpoints, 6,297 users each. (a) Confidence minus the fixed seed-42 order in SID Hit@10 per checkpoint (filled: $B = 64$; hollow: $B = 256$), with depth means. (b) Final beam at $B = 64$: distinct legal SIDs, SIDs absent from the catalog, and duplicate paths to a SID already held; C: confidence, F: fixed. (c) Users who receive ten distinct legal SIDs. “Archived” is the historical decoder, which our search reproduces exactly. Results are conditional on confidence-selected checkpoints and exclude training-seed variation.

(Table 6, Figure 8), and by 0.25 of item-uniform Hit@10 (interval $[+0.15, +0.35]$). All six per-checkpoint differences

Table 5: Verified Industrial RQ-KMeans 3×128 catalog intervention. These are assignment properties, not recommender performance.

Quantity	Native	ZCR
Unique SIDs	2,195	3,105
Items in collision groups	1,391	0
Changed tails	—	910
Final-stage squared distortion	2,322,119	2,428,887

Table 6: Controlled decoding contrasts: confidence minus comparator, in percentage points, with 95% paired-user bootstrap intervals (2,000 draws). Seed-42 rows: 18 retained checkpoints. Six-orders row: mean over all fixed orders on six depth-three checkpoints. Intervals exclude training-seed variation.

Comparator	B	SID Hit@10	NDCG@10
Seed-42	64	-0.95 [-1.07, -0.83]	-0.12 [-0.19, -0.06]
	256	+0.22 [+0.12, +0.34]	+0.31 [+0.25, +0.37]
Six orders	64	+0.40 [+0.29, +0.52]	+0.29 [+0.23, +0.35]

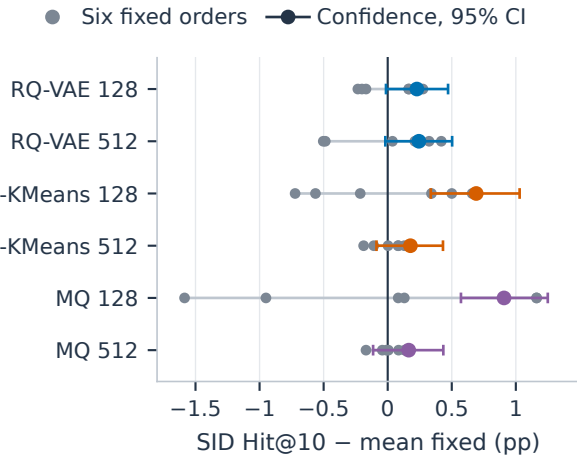


Figure 8: Confidence decoding against all six fixed reveal orders on six depth-three Diff checkpoints at $B = 64$, relative to the mean fixed order. Bars are 95% paired-user intervals. Selecting the best order on test data is a descriptive oracle.

are positive, but four of their intervals include zero. The best fixed order depends on the quantizer. Revealing the last digit first is best for MQ at width 128 and worst for RQ-VAE at both widths, and the best RQ-KMeans orders reveal the second digit first. On three checkpoints the best fixed order scores above confidence, by at most 0.25 points; because it is chosen on test data, it is a descriptive oracle, not a deployable policy. At this depth, duplicate paths still make up 57% of the confidence policy’s final beam.

Interpretation limits. Training determines which partial states the checkpoints have seen. In the vendored implementation that trained them, each example follows one reveal order, ranked by the model’s own confidence on the fully masked target, and depth-five models are never supervised on the fully masked target from which every decode starts. A fixed order can therefore visit partial states that training rarely covers. The results compare search rules on these checkpoints; they are not averaged over training seeds or checkpoint selection rules. Wall time was similar at equal caps (12.4 and 11.6 GPU minutes for confidence and fixed search at $B = 256$ across all 18 checkpoints); the number of distinct candidates was not.

6.7 What the Identifier Maps Encode

We analyzed all 27 Industrial maps against the item embeddings and the Amazon-2018 category hierarchy (Section 5; Figure 9). The results describe the maps, not what trained models use.

Residual digits refine; parallel digits repeat. In the 4×128 maps, the first digit alone explains 53% (RQ-KMeans) and 34% (RQ-VAE) of embedding variance, and later digits 8–18%. Each MQ digit explains 36–38%. Category AMI splits the same way: 0.52 and 0.50 for the residual first digits, against 0.32–0.34 for every MQ digit. Given the earlier digits, MQ’s second digit adds only 0.19 of its 0.33 (0.17–0.23 across the nine MQ maps), whereas RQ-KMeans’s second digit adds 0.14 both alone and conditionally. MQ digits are overlapping partitions of similar quality, so an MQ prefix is an intersection of partitions, not a coarse-to-fine path.

Categories saturate early. Majority-category purity exceeds the shuffled-label null at every depth. At depth three it reaches 0.97 (RQ-KMeans), 0.99 (RQ-VAE), and 0.81 (MQ), against nulls of 0.45, 0.50, and 0.39. Further digits mainly separate items within a category. Groups also tighten faster than chance: in the 5×128 maps the refinement index rises from 0.51 to 0.95 over five RQ-KMeans digits and from 0.35 to 0.80 for MQ.

Where the address budget goes. RQ-KMeans uses every code of every codebook, yet 44.8% of its 3×128 items share a complete SID (MQ: 54.5%). RQ-VAE uses only 2–31% of its first codebook in eight of nine maps (81% at 3×128), including 71 of 256 codes in the planned 3×256 anchor. Its four- and five-digit maps spend the last digit on identity: after four digits of the 4×128 map, 99.2% of RQ-VAE items are alone, against 70.7% (RQ-KMeans) and 63.0% (MQ). The constructions therefore differ in where semantics and identity sit, not only in collision rates. The decoding results are suggestive at most: revealing the last, identity-like RQ-VAE digit first is its worst fixed order at both widths.

A provenance anomaly. At each width, the RQ-KMeans depths are truncations of one fit (every three-digit SID is a prefix of the four-digit SID for all 3,105 items), with one exception: the first three digits of the 5×512 map agree with

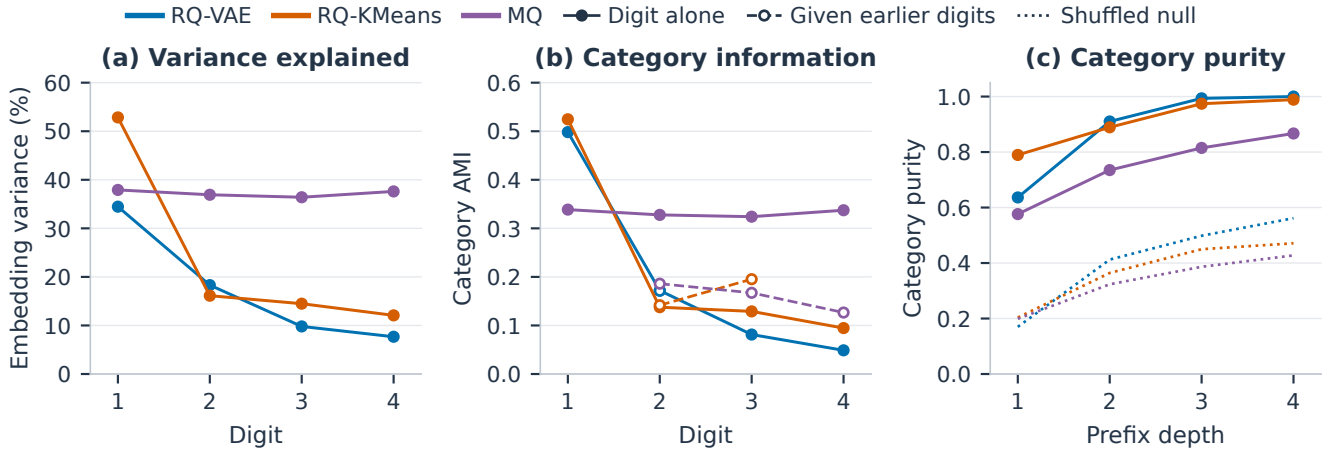


Figure 9: What each digit encodes in the three Industrial 4×128 maps. (a) Item-embedding variance explained by one digit alone. (b) Level-1 category AMI of one digit alone (filled) and given the earlier digits (hollow; shown only where the within-parent estimate is supported). (c) Majority-category purity of multi-item prefix groups (solid) against a shuffled-label null (dotted). At depth four only 0.8% of RQ-VAE items still share a group.

its 3×512 sibling for none of the 3,105 items. It is also the only one of the 27 maps that assigns distinct SIDs to the 11 item pairs with byte-identical embeddings; the other 26 maps collide all 11. Separating identical inputs requires a step beyond quantizing these embeddings, so this map was not produced by the same procedure as its siblings. Its 0.64% collision rate in Table 4 should not be compared with the other RQ-KMeans rows until its construction is documented.

6.8 Static Evidence and Missing Results

Static analyses cover 27 Industrial maps and 504 visible-digit subsets. For RQ-VAE 4×128 , three visible digits determine the remaining digit for 92.54% of catalog items, averaged uniformly over subsets; the corresponding RQ-KMeans 3×128 two-visible-digit value is 33.08%. This describes catalog ambiguity, not learned history use. Geometry summaries and a synthetic slot-swap check, detailed below, likewise do not supply a causal model-intervention result.

Geometry, support, and structural checks. The available shared-space geometry archive contains 135 depth summaries, 108 utilization rows, 432 size-breakdown rows, and 27 collision summaries. RQ-VAE’s original fitted latent space is unavailable. Shrinking cluster dispersion under finer partitions is insufficient evidence of semantic organization without suitable nulls. Section 6.7 reports the semantic-atlas results against permutation and random-partition nulls. The catalog-support pilot contains all 504 subsets and 135 subset-count summaries; these use uniform catalog and subset weights, not empirical training-mask frequencies. For two trained Diff pair checkpoints, a synthetic revealed-slot swap changes masked logits by at most 2.03×10^{-6} and 2.38×10^{-6} , below the check’s 10^{-5} tolerance. This is consistent with revealed-slot

symmetry on synthetic encoder states, not evidence of a desirable property or a real-history retrieval effect.

Matched quality and cost: pending. The compact-system accuracy, common-hardware latency, decoder-call count, and memory comparison has no completed eligible result.

Prediction types: pending. Joint-two, next-only rollout, and horizon-conditioned independent prediction require shared holdout origins and consistent tuple/slot evaluation.

Downstream ZCR: pending. The verified assignment has not yet produced a full native-versus-ZCR recommendation comparison; Table 5 reports geometry only.

Decoding follow-ups: pending. Partial-state deduplication before pruning, the six-order comparison at $B = 256$, and traces of the reveal orders the confidence policy chooses remain to be run.

Neural history use: pending. Matched-history perturbation and activation-intervention results remain unavailable. For the three retained AR checkpoints, prompt reconstruction with a label for every token position and residual-stream capture are implemented. They pass checkpoint-hash, archive-alignment, and batch-invariance checks on an eight-example execution check and contribute no result. Small training/reload/evaluation checks validate execution only and do not occupy missing performance cells.

Replication: pending. We will repeat the matched diagnostic on the prepared Office data. An optional external extension can reuse supplied codes from OpenOneRec/RecIF, subject to adaptation of its task and split definitions.¹ No

¹<https://huggingface.co/datasets/OpenOneRec/OpenOneRec-RecIF>

matched Office or external-dataset result is currently available.

7 Discussion and Conclusion

What the current evidence establishes. The archive makes several dependencies visible: reported quality changes with the evaluation endpoint, item ambiguity changes configuration rankings, and resource measurements depend on training and decoding choices. These findings justify evaluating a system together with its representation, prediction task and resource budget. They do not yet establish which generator is preferable under matched conditions. The existing Qwen and DiffGRM systems remain useful practical references, while the compact control is needed for narrower attribution.

Search is part of the system. Controlled re-decoding adds a direct but narrow intervention result. With the checkpoints fixed, the ordering of two reveal policies reverses between beam caps of 64 and 256. The reversal is largest for five-digit identifiers, where duplicate paths fill most of the confidence beam and the archived decoder returns short lists. Comparisons of generation strategies must therefore report search budgets as distinct candidates or decoder work, not only as beam caps. The identifier maps also differ in where semantics and identity sit. Residual quantizers concentrate standalone category information in the first digit, and parallel MQ digits repeat it. This distinction matters when history and target digits are interpreted in the pending neural analyses.

Limits and pending synthesis. Historical single runs cannot establish seed-robust superiority. Heterogeneous splits and graded pair metrics limit cross-system comparisons; validation/test target overlap additionally restricts historical diffusion pair results. Static code geometry describes available information, not a learned neural mechanism, and it is measured in a shared embedding space rather than in RQ-VAE's unavailable latent space. The decoding results are conditional on single-seed, confidence-selected checkpoints and one catalog. The planned intervention study must distinguish a change in model response from search pruning and distinguish causal internal use from probe decodability. Final recommendations about prediction formulation, inference budget and history use will be added after the matched experiments and their replications are complete.

Data and scope. The empirical scope is sequential catalog recommendation in the retained commerce datasets. Offline interaction prediction does not measure the causal effect of recommendations on users, and edited histories are model-input interventions rather than counterfactual human preferences. Analyses should retain only the item/user identifiers needed for reproducibility and avoid identifying individual reviewers. Upstream data provenance and release terms must be verified before a public artifact release. Conclusions about other domains or conversational recommendation require additional evidence.

Conclusion. The completed analyses show why a useful comparison of generative recommenders must account for system resources, search budgets, prediction semantics and item identity. They provide a foundation for the matched AR/masked-generation diagnostic and focused history analysis. A final conclusion about their relative quality and cost is intentionally deferred until those results are available.

References

- [1] Hongtao Huang, Chengkai Huang, Junda Wu, Tong Yu, Julian McAuley, and Lina Yao. 2025. Listwise Preference Diffusion Optimization for User Behavior Trajectories Prediction. In *Advances in Neural Information Processing Systems*, Vol. 38. Curran Associates, Inc., 159383–159408. doi:10.52202/085713-5324
- [2] Clark Mingxuan Ju, Liam Collins, Leonardo Neves, Bhuvish Kumar, Louis Yufeng Wang, Tong Zhao, and Neil Shah. 2025. Generative Recommendation with Semantic IDs: A Practitioner's Handbook. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25)*. Association for Computing Machinery, 6420–6425. doi:10.1145/3746252.3761612
- [3] Dueun Kim and Albert No. 2026. The Confidence Shortcut: A Reasoning Failure Mode of Masked Diffusion Models. arXiv:2605.29123 [cs.AI] doi:10.48550/arXiv.2605.29123
- [4] Anton Klenitskiy, Konstantin Polev, Daria Denisova, Alexey Vasilev, Dmitry Simakov, and Gleb Gusev. 2025. Sparse Autoencoders for Sequential Recommendation Models: Interpretation and Flexible Control. arXiv:2507.12202 [cs.IR] doi:10.48550/arXiv.2507.12202 Revised February 2026.
- [5] Injin Kong, Hyoungjoon Lee, and Yohan Jo. 2026. Mechanism Shift During Post-training from Autoregressive to Masked Diffusion Language Models. arXiv:2601.14758 [cs.LG] doi:10.48550/arXiv.2601.14758 Version 5, August 2026; accepted at EMNLP 2026.
- [6] Xiaoyu Kong, Leheng Sheng, Junfei Tan, Yuxin Chen, Jiancan Wu, An Zhang, Xiang Wang, and Xiangnan He. 2025. MiniOneRec: An Open-Source Framework for Scaling Generative Recommendation. arXiv:2510.24431 [cs.IR] doi:10.48550/arXiv.2510.24431
- [7] Qijiong Liu, Jieming Zhu, Lu Fan, Kun Wang, Hengchang Hu, Wei Guo, Yong Liu, and Xiao-Ming Wu. 2025. Can LLMs Outshine Conventional Recommenders? A Comparative Evaluation. In *Advances in Neural Information Processing Systems*, Vol. 38. Curran Associates, Inc. doi:10.52202/085713-0424
- [8] Zhao Liu, Yichen Zhu, Yiqing Yang, Xiao Lv, Guoping Tang, Rui Huang, Qiang Luo, Ruiming Tang, Kun Gai, and Guorui Zhou. 2026. DiffGRM: Diffusion-based Generative Recommendation Model. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*. Association for Computing Machinery, 5853–5864. doi:10.1145/3774904.3792156
- [9] Shashank Rajput, Nikhil Mehta, Anima Singh, Raghunandan H. Keshavan, Trung Vu, Lukasz Heldt, Lichan Hong, Yi Tay, Vinh Q. Tran, Jonah Samost, Maciej Kula, Ed H. Chi, and Maheswaran Sathiamoorthy. 2023. Recommender Systems with Generative Retrieval. In *Advances in Neural Information Processing Systems*, Vol. 36. Curran Associates, Inc., 10299–10315. doi:10.52202/075280-0452
- [10] Kulin Shah, Bhuvish Kumar, Neil Shah, and Liam Collins. 2025. Masked Diffusion for Generative Recommendation. arXiv:2511.23021 [cs.LG] doi:10.48550/arXiv.2511.23021
- [11] Tuan-Binh Tran, Thanh Tam Nguyen, Quoc Viet Hung Nguyen, Dung D. Le, Tung Kieu, and Thanh Trung Huynh. 2026. EPIC: Explicit Posterior Item Conditioning for Semantic ID Diffusion Recommendation. arXiv:2609.03522 [cs.IR] https://arxiv.org/abs/2609.03522
- [12] Junting Wang, Xinrui He, Yunzhe Li, and Hari Sundaram. 2026. Understanding Semantic IDs: From Item Representation to Item Selection in Generative Recommendation. arXiv:2607.24995 [cs.AI] doi:10.48550/arXiv.2607.24995
- [13] Xu Wang, Bingqing Jiang, Yu Wan, Baosong Yang, Lingpeng Kong, and Difan Zou. 2026. DLM-Scope: Mechanistic Interpretability of Diffusion Language Models via Sparse Autoencoders. arXiv:2602.05859 [cs.LG] doi:10.48550/arXiv.2602.05859
- [14] Qian Zhang, Lech Szymanski, Jeremiah D. Deng, and Haibo Zhang. 2026. Faithful Evaluation of Semantic-ID Tokenizers for Generative Recommendation. In *Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM '26)*. Association for Computing Machinery, New York, NY, USA, 12 pages. doi:10.1145/3799682.3841124
- [15] Zhen Zhang, Jujia Zhao, Xinyu Ma, Xin Xin, Maarten de Rijke, and Zhaochun Ren. 2026. Cold-Starts in Generative Recommendation: A Reproducibility Study. arXiv:2603.29845 [cs.IR] doi:10.48550/arXiv.2603.29845